

SKIN AND HAIR DEFENSIVE FUNCTIONS

Skin is the body's first line of defence against potential harm. As such, it is well equipped to prevent physical damage due to its supple, cushioned qualities. The epidermal cells that form skin's outermost layers are tightly knit together, but allow a certain amount of pliability. The cells are almost entirely full of the tough protein keratin, which resists attack by many kinds of chemicals. The natural secretion of sebum from the millions of sebaceous glands, each associated with a hair follicle, is slightly oily at body temperature and spreads easily. It furnishes the skin with partially water-repellent and antibiotic qualities, inhibiting the growth of certain microorganisms, and prevents hairs from becoming too brittle.

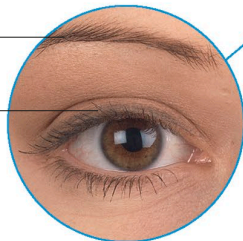
SCALP HAIR

Head hairs help to keep rainwater from the scalp, absorb or deflect some of the energy in knocks and blows, and shield the head from extremes of temperature.



eyebrow

eyelash



Sebum oils and waxes

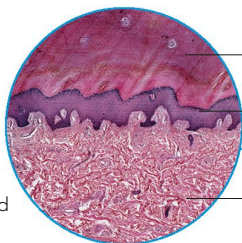
Mixture of lipid-rich secretions (palmitic, stearic, oleic, linoleic, and other fatty acids) softens and lubricates skin and repels water

EYEBROWS AND EYELASHES

The arch of relatively coarse, fast-growing eyebrow hairs helps to divert sweat or rainwater on the forehead that might trickle into the eyes. Eyelashes produce swirling air currents when blinking, which push floating particles away from the eye surface.

THICKENED SKIN

Areas of skin subjected to regular pressure respond by thickening their epidermis for greater protection and buffering, as in this magnified image of skin from the foot.



Thick epidermis

Basal cell layer

Dermis

Toenails

Made of almost solid keratin

